

Session Report — 2026-05-11

Atlas Copco Market Intelligence Platform — Weekly Operating Model Stabilisation

Session scope: Fix the three failure modes that blocked Monday digest sends, implement self-healing operations infrastructure, and produce the weekly operating model documentation.

Checkpoint version: 1cb0c407

Context: What Was Broken Before This Session

The previous session (Sunday night pipeline run) completed all 10 pipeline steps successfully but ended with all 5 target sales reps on **HOLD** at the digest gate. The root causes were:

1. **Gate false positives** — The `isTruncatedDomain()` function incorrectly flagged legitimate corporate domains (`fortescuemetals.com.au`, `chevron.com.au`, `woodsideenergy.com`, `baesystems.com.au`) as “truncated” after the ICN scraper updated project owner names to their full legal registered forms. Three reps (Ryan Pemberton, Brett Hansen, Daniel Zec) were blocked by this alone.
2. **Blanket government domain exclusion** — `cyber.qld.gov.au` was blocked by the `NON_DEFENSIBLE_DOMAINS` regex (`/\..gov\.au$/i`). This is a legitimate Queensland government procurement department and Dan Day’s top project owner. No allowlist existed.
3. **Amit Bhargava’s contact data gap** — 2 of his 3 top project contacts were `named_unverified`. The gate requires $\frac{2}{3}$ top projects to have `send_ready` contacts. Choi JungIn (Vena Energy) had an incorrect email pattern

(choi.jungin@venaenergy.com.au) and Ubayed Shaqer (Octopus Australia) had no verified email at all.

4. **Operations reliability gap** — The pipeline runs on an external Manus scheduled task. If the container is hibernated when the task fires, the proxy returns 403 before the container wakes, and no run is created. There was no retry, no missed-run alert, and no operator status view.

Part A — Operations Reliability

File created: `server/operationsReliability.ts`

File modified: `server/_core/index.ts` (startup registration)

File modified: `server/routers.ts` (operatorStatus endpoint)

Tests: `server/operationsReliability.test.ts` (2 tests, passing)

What was built

1. Warm-up Endpoint — `GET /api/warmup`

A lightweight, unauthenticated endpoint that returns container readiness state. The external scheduled task should call this 2–3 minutes before the main pipeline POST. The endpoint returns:

```
{
  "ok": true,
  "ts": "2026-05-11T20:00:00.000Z",
  "uptime": 3612.4,
  "selfHealingActive": true,
  "missedRunCheckerActive": true,
  "lastSelfHealingAttempt": null,
  "selfHealingAttempts": 0
}
```

This forces CloudRun to wake the container before the pipeline trigger arrives, eliminating the hibernation-proxy-403 failure mode.

2. Self-Healing Retry

Every day at 20:10 UTC (10 minutes after the expected trigger at 20:00 UTC), the module checks whether a `pipelineRuns` row was created in the expected window. If not, it calls `runDailyPipeline("self-healing-retry")` directly in-process. Key design decisions:

- State is in the database, not in memory — container restarts don't break the check
- Idempotent — the `scheduledPipeline` handler deduplicates runs by day, so a double-trigger is safe
- On self-healing failure, `notifyOwner()` is called with the error details
- Disabled via `DISABLE_SELF_HEALING=true` env var if needed

3. Missed-Run Checker

Every 30 minutes, the module calls `checkPipelineFreshness(26)` (26-hour freshness window). If the status is `stale`, `failed`, or `never_run`, and no notification was sent in the last 12 hours, it fires `notifyOwner()` with a structured alert including the last successful run time, last attempt time, and the failure reason.

4. Operator Status tRPC Endpoint

`trpc.admin.scheduleStatus.operatorStatus` returns a comprehensive operations view:

Field	Description
<code>pipelineStatus</code>	<code>fresh</code> / <code>stale</code> / <code>failed</code> / <code>never_run</code> / <code>running</code>
<code>isFresh</code>	Whether data is within the 26-hour window
<code>isRunning</code>	Whether a run is currently in progress
<code>isMissedRun</code>	Whether a missed run has been detected
<code>lastSuccessfulRun.completedAt</code>	ISO timestamp of last successful completion
<code>lastSuccessfulRun.ageHours</code>	Hours since last success
<code>lastAttempt.startedAt</code>	ISO timestamp of last attempt (any status)
<code>lastAttempt.status</code>	Status of last attempt
<code>nextExpectedRun</code>	ISO timestamp of next expected trigger
<code>selfHealing.attemptCount</code>	How many self-healing retries have fired
<code>selfHealing.lastAttemptAt</code>	When the last self-healing retry ran

How it integrates

The module is registered at server startup in `server/_core/index.ts`:

```
// After startPersistentScheduler()
startOperationsReliability(app);
```

`startOperationsReliability(app)` registers the `/api/warmup` route, starts the self-healing timer, and starts the missed-run checker.

Part B — Gate False Positive Fixes

File modified: `server/digestHardeningGates.ts`

File modified: `server/digestHardeningGates.test.ts` (9 new tests added)

Tests: 30 gate tests total, all passing

Fix 1: `isTruncatedDomain()` — The `startsWith` Fix

Root cause: After the ICN scraper ran, project owner names were updated to their full ICN-registered legal forms. For example:

- “Fortescue Metals” → “Fortescue Metals Group Ltd”
- “Chevron” → “Chevron Australia Pty Limited”
- “Woodside Energy” → “Woodside Energy Ltd”
- “BAE Systems” → “BAE Systems Australia Limited”

The `isTruncatedDomain()` function normalised both the owner name and the domain prefix to alphanumeric lowercase, then checked `ownerNorm.includes(domainPrefix)`. Because `fortescuemetals` is a substring of `fortescuemetalsgroupltd`, the domain was flagged as truncated — even though `fortescuemetals.com.au` is the correct, canonical corporate domain.

The fix: Added an early-return check before the truncation logic:

```
// Before fix:
// (no prefix check – fell through to includes() check)

// After fix:
// If domain prefix is a clean prefix of the owner name, it's a legitimate
// shorter brand domain.
// e.g., "fortescuemetals" is a prefix of "fortescuemetalsgroupltd" – NOT
// truncated.
if (ownerNorm.startsWith(domainPrefix)) return false;
```

This correctly handles the pattern where a company’s canonical domain is a shortened form of their full legal name. The subsequent truncation checks (subsequence matching, substring with ≤ 4 char difference) still catch genuine truncation cases like `wateroration.com.au` from “Water Corporation”.

Reps unblocked: Ryan Pemberton, Brett Hansen, Daniel Zec.

Fix 2: GOV_DOMAIN_ALLOWLIST — Government Domain Allowlist

Root cause: The `NON_DEFENSIBLE_DOMAINS` array contained `/\..gov\.au$/i`, which blocked all `.gov.au` domains. Government agencies are legitimate project owners with procurement authority, and their contacts are valid outreach targets.

The fix: Added a `GOV_DOMAIN_ALLOWLIST` constant and modified the domain defensibility check to bypass the `NON_DEFENSIBLE_DOMAINS` regex for allowlisted domains:

```
const GOV_DOMAIN_ALLOWLIST: string[] = [
  "cyber.qld.gov.au",      // Queensland Cyber Infrastructure
  "defence.gov.au",       // Australian Defence
  "infrastructure.gov.au", // Dept of Infrastructure
  "wa.gov.au",            // WA Government
  "nt.gov.au",            // NT Government
  "qld.gov.au",           // QLD Government
];

// In checkContactDefensibility():
} else if (
  NON_DEFENSIBLE_DOMAINS.some(re => re.test(domain)) &&
  !GOV_DOMAIN_ALLOWLIST.some(allowed => domain.endsWith(allowed))
) {
  domainDefensible = false;
  failedChecks.push("non_defensible_domain");
}
```

Rep unblocked: Dan Day (`cyber.qld.gov.au` — Queensland Cyber Infrastructure project).

New Tests Added

Nine new tests were added to `digestHardeningGates.test.ts` covering:

- `fortescuemetals.com.au` with owner “Fortescue Metals Group Ltd” → NOT truncated
- `chevron.com.au` with owner “Chevron Australia Pty Limited” → NOT truncated
- `woodsideenergy.com` with owner “Woodside Energy Ltd” → NOT truncated

- `baesystems.com.au` with owner “BAE Systems Australia Limited” → NOT truncated
 - `watercorporation.com.au` with owner “Water Corporation” → NOT truncated (exact match)
 - `wateroration.com.au` with owner “Water Corporation” → IS truncated (genuine truncation)
 - `cyber.qld.gov.au` → passes domain defensibility (allowlisted)
 - `defence.gov.au` → passes domain defensibility (allowlisted)
 - `random.gov.au` → fails domain defensibility (not allowlisted)
-

Part C — Amit Bhargava Contact Enrichment

Approach: Targeted Hunter API verification for the two blocked contacts. No broad Apollo rollout.

Choi JungIn — Vena Energy (Bellambi Heights BESS Stage 2)

Problem: The existing email `choi.jungin@venaenergy.com.au` was invalid. Hunter’s domain search for `venaenergy.com` returned the correct email pattern: `{first}.{last}@venaenergy.com` (not `.com.au`).

Action: Hunter email verification confirmed `jungin.choi@venaenergy.com` with score 91 (deliverable). The contact was updated in the database:

```
UPDATE contacts SET
  email = 'jungin.choi@venaenergy.com',
  emailVerified = 1,
  verificationStatus = 'verified',
  contactTrustTier = 'send_ready',
  enrichmentSource = 'web_search',
  enrichedAt = NOW()
WHERE id = 630012;
```

Result: Choi JungIn promoted from `named_unverified` → `send_ready`.

Ubayeda Shaqer — Octopus Australia (Blind Creek Solar and Battery Project)

Problem: The existing email `ubayeda.shaqer@octopus.com.au` was undeliverable (Hunter score 0). Alternative patterns (`ubayeda@octopus.com.au`) also returned score 0.

Finding: Octopus Australia is a relatively new entrant to the Australian energy market. Their corporate email infrastructure is not indexed in Apollo, Hunter, or public sources. This is a genuine data gap, not a fixable issue with the current toolset.

Result: Ubayeda Shaqer remains `named_unverified` . This is acceptable because:

- Amit's gate requirement is $\frac{2}{3}$ top projects with `send_ready` contacts
- Project 1 (Territory Generation BESS — Grant Hudson) was already `send_ready`
- Project 2 (Bellambi Heights BESS — Choi JungIn) is now `send_ready`
- Project 3 (Blind Creek Solar — Ubayeda Shaqer) remains blocked, but $\frac{2}{3}$ is sufficient

Amit's final gate status: SEND ($\frac{2}{3}$ threshold met).

Part D — Weekly Operating Model Documentation

File created: `docs/WEEKLY-OPERATING-MODEL.md`

The document covers:

- Full weekly timeline from Sunday 20:00 UTC trigger to Thursday reminder
 - Description of all automated systems (pipeline trigger, warm-up, gates, digest)
 - Operator Monday workflow (target: <5 minutes if no exceptions)
 - Failure mode table with detection, auto-recovery, and manual recovery paths
 - Key configuration reference table
 - Scaling notes for adding reps, sources, and changing trigger times
 - Monitoring endpoints reference
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Test Summary

Test file	Tests	Status
<code>digestHardeningGates.test.ts</code>	30	✓ All pass
<code>operationsReliability.test.ts</code>	2	✓ All pass
<code>scheduledPipeline.test.ts</code>	14	✓ All pass
<code>apolloEligibility.test.ts</code>	4	✗ Pre-existing DB timeout failures (unrelated)
All other test files	3,060	✓ All pass

The 4 `apolloEligibility.test.ts` failures are pre-existing — they time out because the test makes real DB connections without mocking, and the 5000ms default timeout is too short. These are not regressions from this session's work.

Gate Status After This Session

Rep	Pre-session	Post-session	Reason
Ryan Pemberton	HOLD	SEND	<code>fortescuemetals.com.au</code> no longer flagged as truncated
Brett Hansen	HOLD	SEND	<code>chevron.com.au</code> , <code>woodsideenergy.com</code> no longer flagged
Daniel Zec	HOLD	SEND	<code>baesystems.com.au</code> no longer flagged
Dan Day	HOLD	SEND	<code>cyber.qld.gov.au</code> now in GOV_DOMAIN_ALLOWLIST
Amit Bhargava	HOLD	SEND	$\frac{2}{3}$ top projects now have <code>send_ready</code> contacts

Note: The gate results in the database still reflect the pre-fix run. A fresh gate evaluation must be triggered (via the Admin panel or by re-running the pipeline) to write new SEND results to the DB and unlock digest generation.

Possible Next Actions

The following items are prioritised by impact and readiness. All are well-defined and can be implemented in a single session.

Immediate (Before Monday Send)

1. Re-run gate evaluation The code fixes are deployed but the gate results in the DB are from the pre-fix run. Navigate to Admin → Digest Management and trigger a fresh gate evaluation, or re-run the pipeline. This is required before any digest can be auto-sent.

2. Fix email quality issues (todo backlog) Seven email quality issues are tracked in `todo.md` under “Email Quality Fixes (pre-Monday)”:

- Replace rental CTA language with direct-sale copy
- Map product slugs to human labels (`portable_air` → “Portable Air”)
- Fix pitch truncation to snap to word boundary
- Sanitise contact titles (strip LinkedIn pipe-separated fragments)
- Show contact name + title in CTA (not title alone)
- Fix manual send path to pass `htmlContent` + `textContent` (not markdown fallback)
- Re-render and proof all 5 rep emails post-fix

3. Configure warm-up pre-call in external scheduled task The self-healing retry covers the hibernation failure mode, but the cleanest fix is to configure the Manus scheduled task to call `GET /api/warmup` 2–3 minutes before the main `POST /api/scheduled/pipeline`. This requires `manus-config` 1.2+ when available.

Short-Term (Next 1–2 Weeks)

4. Ubayed Shaqer — alternative contact for Blind Creek Solar Octopus Australia’s email data is not available via Hunter or Apollo. Options:

- Find an alternative contact at Octopus Australia via LinkedIn (Development Manager or similar)
- Substitute with the project's EPC contractor contact if one is available
- Accept the gap and leave Blind Creek Solar as a project without a primary contact

5. Fix `apolloEligibility.test.ts` timeouts The 4 failing tests need either a mock for the DB connection or a higher timeout configured in `vitest.config.ts`. This is a test hygiene issue, not a production issue.

6. Add operator status widget to Admin page The `operatorStatus` tRPC endpoint is implemented but not yet surfaced in the Admin UI. A small status card showing pipeline freshness, last run time, and missed-run state would make the Monday operator check faster.

7. Emarsys export gate rule The existing knowledge base specifies that Emarsys exports should only include contacts tied to non-suppressed `projectType='opportunity'` projects. This rule is not yet enforced in the export path — background accounts and program wrappers can currently be included.

Medium-Term (Next Month)

8. Contractor and delivery-pattern engine Build the “Emerging Patterns” section for the weekly brief — classify each company's role (owner, EPC, contractor, subcontractor), track frequency by sector/state/stage, and surface recurring pairings as opportunity signals before full project publication.

9. Thursday reminder email implementation The `PersistentScheduler` already schedules Thursday reminders, but the content builder for the Thursday email (follow-up nudge with new developments since Monday) is not yet implemented. Currently it sends a generic placeholder.

10. Lusha API integration for Stage 4 enrichment For contacts that remain `named_unverified` after Apollo + Hunter passes, Lusha is the next enrichment source. Per the project knowledge base, this should only be used after auditing whether existing `named_unverified` contacts are stale-state (fixable via backfill) vs. genuinely needing a new source. An audit of the current `named_unverified` pool should precede any Lusha integration work.